From Förster transfer to directed percolation: the photoswitching dye lattice as a contact process

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Abstract

A lattice of photoswitching Cy5 dyes, in which an ON dye can return a dark neighbour to ON by energy transfer, realises a contact process. We derive the coupling between two dyes from Förster theory and the dye's switching photophysics, show that the resulting dynamics satisfies the conditions of the directed-percolation (DP) conjecture, and treat the three experimental imperfections — spontaneous recovery, photobleaching and quenched disorder — as perturbations of the DP fixed point. Each perturbation gives a crossover time that scales as a power of its strength. The predicted crossover times agree with exact stochastic simulations to within about 20 %. The scaling arguments are ours; the DP exponents and the Harris criterion are standard.

1 Microscopic model

Each dye i is in one of three states: ON ($n_i = 1$), dark but recoverable ($n_i = 0$), or bleached. An ON dye is excited at rate k_{exc}, proportional to the laser irradiance. From its excited singlet state S_1 the dye can

1. relax to the ground state, by fluorescence or internal conversion (total rate $k_D = 1/\tau_D$);
2. enter the long-lived dark state, by photoinduced electron transfer from thiolate, either directly or via the triplet; we write the probability per excitation as p_{off};
3. transfer its energy to a dark neighbour j at the Förster rate

$$k_T(d_{ij}) = k_D \left(\frac{R_0}{d_{ij}}\right)^6, \qquad R_0^6 = \frac{9 \ln 10\, \kappa^2 \Phi_D}{128 \pi^5 n^4 N_A} J_{\mathrm{off}}, \tag{1}$$

   where J_{off} is the overlap of the ON-dye emission with the dark-state absorption. A transfer returns j to ON with probability η.

Bleaching ends an ON period with probability p_{b}. Figure 1 summarises the rules.

Rates. Let $E_{ij} = k_T(d_{ij})/(k_D + \sum_l k_T(d_{il}))$ be the fraction of i's excitations transferred to j. The rate at which ON dye i wakes dark dye j, and the rate at which i itself goes dark, are

$$\lambda_{ij} = k_{\mathrm{exc}}\, \eta\, E_{ij}, \qquad k_{\mathrm{off}} = k_{\mathrm{exc}}\, p_{\mathrm{off}}. \tag{2}$$

Both are proportional to k_{exc}, so the excitation rate sets only the clock. Measuring time in ON lifetimes, $t \to k_{\mathrm{off}} t$, the only coupling is

$$r_{ij} = \frac{\lambda_{ij}}{k_{\mathrm{off}}} = \frac{\eta}{p_{\mathrm{off}}}\, E_{ij}. \tag{3}$$

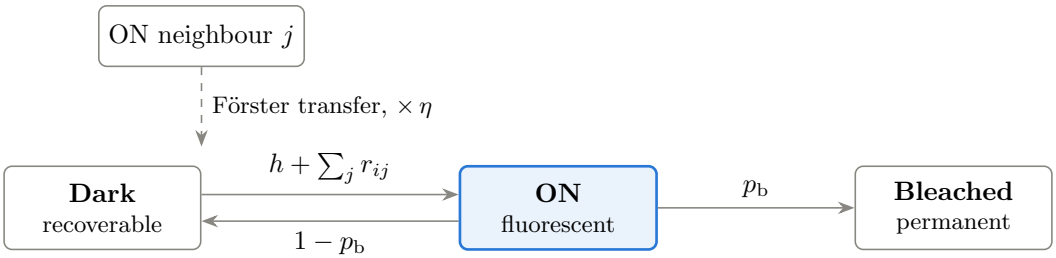


Figure 1: The dye rule, with rates in units of the off rate k_off. Only the recovery rate depends on the neighbours; r_{ij} comes from Förster transfer (Eq. 4).

In the weak-transfer limit, $E_{ij} \approx (R_0/d_{ij})^6$, this becomes

$$\boxed{r(d) = \left(\frac{d_0}{d}\right)^6, \qquad d_0 = R_0 \left(\frac{\eta}{p_\mathrm{off}}\right)^{1/6}} \tag{4}$$

For a pair, the hand-over probability (an ON dye wakes its dark neighbour before going dark) is $B = r/(1+r)$.

Numbers. With the Cy5 spectra and constants of Ebert, Sauer and Doose (2026), $R_0 = 4.29\,\mathrm{nm}$, $\eta = 10^{-4}$ and $p_\mathrm{off} = 6.5 \times 10^{-6}$ (100 mM MEA, pH 7.5), Eq. (4) gives $d_0 = 6.77\,\mathrm{nm}$, in agreement with the full model's 6.74 nm. If the transferred photon wakes the dark dye no more efficiently than a directly absorbed one ($\eta \approx 2 \times 10^{-7}$), then $d_0 = 2.38\,\mathrm{nm}$.

Sensitivity. Eq. (4) has two useful consequences. A critical coupling r_c fixes a critical spacing

$$d_c = d_0\, r_c^{-1/6}, \tag{5}$$

so a factor-of-two error in r_c moves d_c by only 12 %. Conversely, spacing errors are amplified: $\delta r/r = -6\,\delta d/d$. Only p_off depends on thiolate, so pH and thiol concentration tune r at fixed geometry.

2 Mapping onto the contact process

With $p_\mathrm{b} = 0$ and no spontaneous recovery, the master equation is generated by

$$n_i :\ 1 \to 0 \text{ at rate } 1, \qquad n_i :\ 0 \to 1 \text{ at rate } \sum_j r_{ij}\, n_j. \tag{6}$$

This is the contact process with control parameter $r_1 = r(a)$, a the nearest-neighbour distance. The state $n_i \equiv 0$ is absorbing. Below a critical r_c activity dies; above it a finite density of ON dyes persists.

2.1 Conditions of the DP conjecture

Janssen and Grassberger conjectured that a continuous transition into a single absorbing state, with short-range interactions, no additional symmetry or conservation law, and no quenched disorder, belongs to the DP class. We check each condition.

Single absorbing state. Holds without spontaneous recovery. Bleached sites are inert and do not create a second absorbing state.

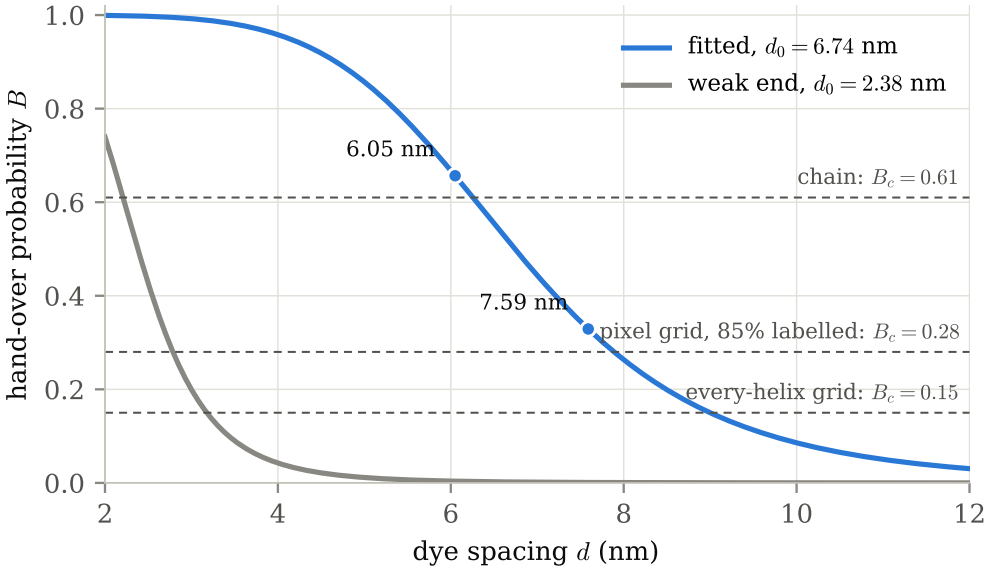


Figure 2: Hand-over probability $B = r/(1+r)$ against spacing for the fitted and weak ends of the coupling. Dashed lines: critical values B_c per nearest bond from simulation (Sec. 4). Dots: the two origami grids under the fitted law.

Range of the interaction. The coupling decays as d^{-6}. For DP with long-range spreading $P(d) \propto d^{-(D+\sigma)}$, the long-range term is irrelevant when σ exceeds a threshold close to 2 in the dimensions of interest. In $D = 2$, $D + \sigma = 6$ gives $\sigma = 4$. Förster coupling is therefore short-range for the purposes of DP.

No extra symmetry or conservation. The rules conserve nothing and have no particle–hole or parity symmetry.

Non-additive rates. When several dark neighbours surround an ON dye, they compete for the same excitation through the denominator of E_{ij}. The wake rates are then sublinear in the number of dark neighbours. In addition, excitations lost to transfer are not available for switching off, so an ON dye with many dark neighbours lives longer. At $d = 6\,\text{nm}$ a single neighbour takes about 11 % of the excitations and four take about 45 %. These are local, state-dependent rates that preserve the absorbing state; they shift r_c but leave the universality class unchanged.

Quenched disorder. Fails in principle; see Sec. 3.3.

2.2 Exponents

We use the standard $2+1$-dimensional DP values:

Exponent	Value	Defined by
β	0.583	$\rho_{\text{st}} \sim (r - r_c)^{\beta}$
$\nu_\perp$	0.733	$\xi_\perp \sim \lvert r - r_c\rvert^{-\nu_\perp}$
$\nu_\parallel$	1.295	$\xi_\parallel \sim \lvert r - r_c\rvert^{-\nu_\parallel}$
$\delta = \beta/\nu_\parallel$	0.4505	$\rho(t) \sim t^{-\delta}$ at r_c, from a fully active start
σ	2.18	field exponent, $\rho_{\text{st}}(r_c, h) \sim h^{\beta/\sigma}$

3 Perturbations of the DP fixed point

Each imperfection is a relevant perturbation. At criticality, it is invisible at short times and takes over after a crossover time that scales as a power of its strength. That time is the length of the observable critical window (Fig. 3).

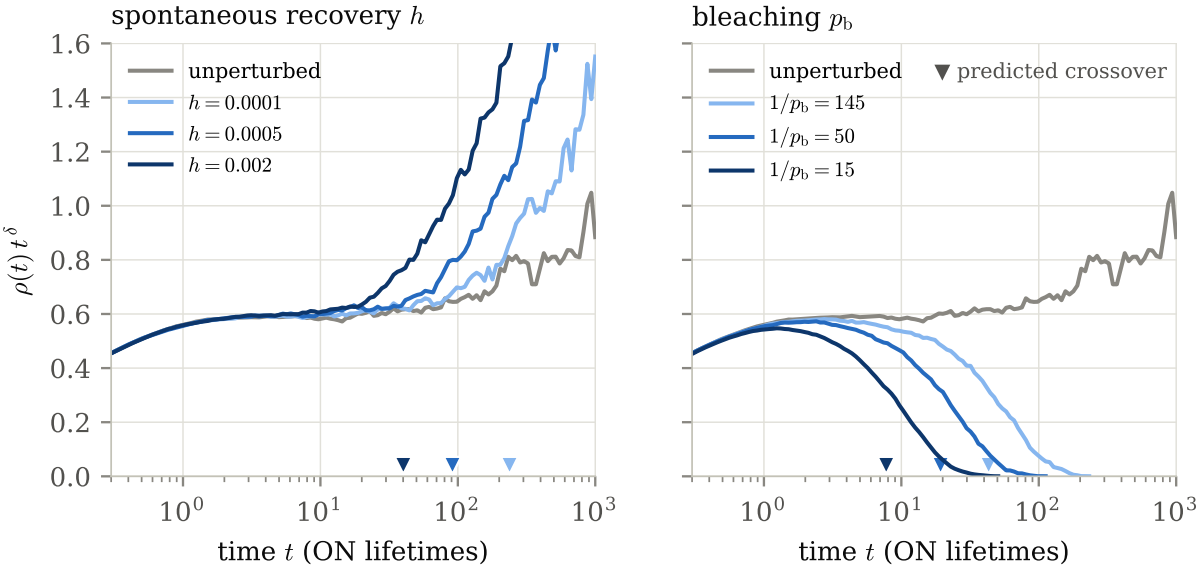


Figure 3: Critical decay at the threshold of the pixel grid ($r_1 = 0.334$, 256×256 sites, two runs each), scaled by t^{δ} so that pure DP is flat. Spontaneous recovery bends the curve up, bleaching bends it down. Triangles: crossover times predicted by Eqs. (8) and (11) with unit prefactor.

3.1 Spontaneous recovery as the conjugate field

Dark dyes recover on their own at rate h (per ON lifetime), giving the extra transition $n_i : 0 \to 1$ at rate h. This is the field conjugate to the order parameter in DP. At $r = r_c$ the scaling form

$$\rho(t, h) = t^{-\delta}\, F\Big(h\, t^{\sigma/\nu_\parallel}\Big) \tag{7}$$

gives a crossover when $h\, t^{\sigma/\nu_\parallel} \sim 1$:

$$\boxed{t_h \sim h^{-\nu_\parallel/\sigma} = h^{-0.59}} \tag{8}$$

3.2 Photobleaching as activity-driven dilution

Each ON period removes a dye with probability p_b, so the density of live dyes $c(t)$ obeys

$$\dot{c} = -p_b\, \rho(t). \tag{9}$$

A lower site density lowers the effective coupling, which acts as a shift of the control parameter proportional to the accumulated activity:

$$\Delta(t) \;\propto\; p_b \int_0^t \rho(t')\, dt' \;\sim\; p_b\, t^{1-\delta}, \tag{10}$$

using $\rho \sim t^{-\delta}$ at criticality. The shift becomes comparable to the distance from criticality probed at time t, $t^{-1/\nu_\parallel}$, when $p_b\, t^{1-\delta+1/\nu_\parallel} \sim 1$:

$$\boxed{t_b \sim p_b^{-1/(1-\delta+1/\nu_\parallel)} = p_b^{-0.76}} \tag{11}$$

The exponent is larger than in Eq. (8), so for comparable strengths bleaching ends the critical window first. This is the immunisation-type perturbation studied by Jiménez-Dalmaroni and Hinrichsen (2003), in which reinfection differs from first infection and the DP fixed point is unstable.

3.3 Quenched disorder

Missing dyes, frozen position errors and frozen dipole orientations make the couplings r_{ij} random but fixed in time. By the Harris criterion, such disorder is relevant at the DP fixed point if

$$D\,\nu_\perp < 2. \tag{12}$$

In $D = 2$, $D\nu_\perp = 1.47$, so disorder is relevant. For strong disorder the transition flows to an infinite-randomness fixed point with activated scaling and a Griffiths region (Vojta, Farquhar and Mast 2009). For weak disorder the crossover length is large, and bleaching ends the critical window first.

Three sources of disorder are specific to the dye lattice:

- *Missing dyes.* Staple incorporation is about 84 %; vacancies raise r_c approximately in proportion to $1/c$.
- *Spacing errors.* Through $\delta r/r = -6\,\delta d/d$, a 1 nm error at 6 nm changes a bond by about 100 %. Because the coupling is convex in d, averaging over spacing errors raises the mean coupling.
- *Orientation factor.* Eq. (1) contains κ^2. If the dyes rotate freely within the excited-state lifetime, $\kappa^2 = 2/3$ for every bond. If they are frozen (Cy5 partly stacks on DNA), κ^2 varies between 0 and 4 from bond to bond. This could be the largest source of disorder, and it is unmeasured.

4 Comparison with simulation

We simulated the rules of Sec. 2 exactly (Gillespie algorithm) on a 256×256 pixel grid at its threshold, $r_1 = 0.334$, quenched from fully ON. Table 1 compares the time at which $\rho(t)$ departs from the unperturbed critical decay with Eqs. (8) and (11) (see also Fig. 4). The prefactors of the scaling forms are set to 1; only the exponents are predicted.

Table 1: Crossover times in ON lifetimes: scaling prediction against simulation.

Perturbation	Strength	Predicted	Simulated
Field, Eq. (8)	$h = 0.002$	40	36 (20 % departure)
	$h = 0.0005$	91	86
	$h = 0.0001$	238	284
Bleaching, Eq. (11)	$1/p_\mathrm{b} = 15$	7.8	9.1 (50 % departure)
	$1/p_\mathrm{b} = 50$	19	21
	$1/p_\mathrm{b} = 145$	43	44

The agreement in scaling across a factor of 20 in h and 10 in p_b supports both the mapping to DP and the treatment of the perturbations. The absolute agreement depends on the choice of departure threshold, which absorbs the unknown prefactor.

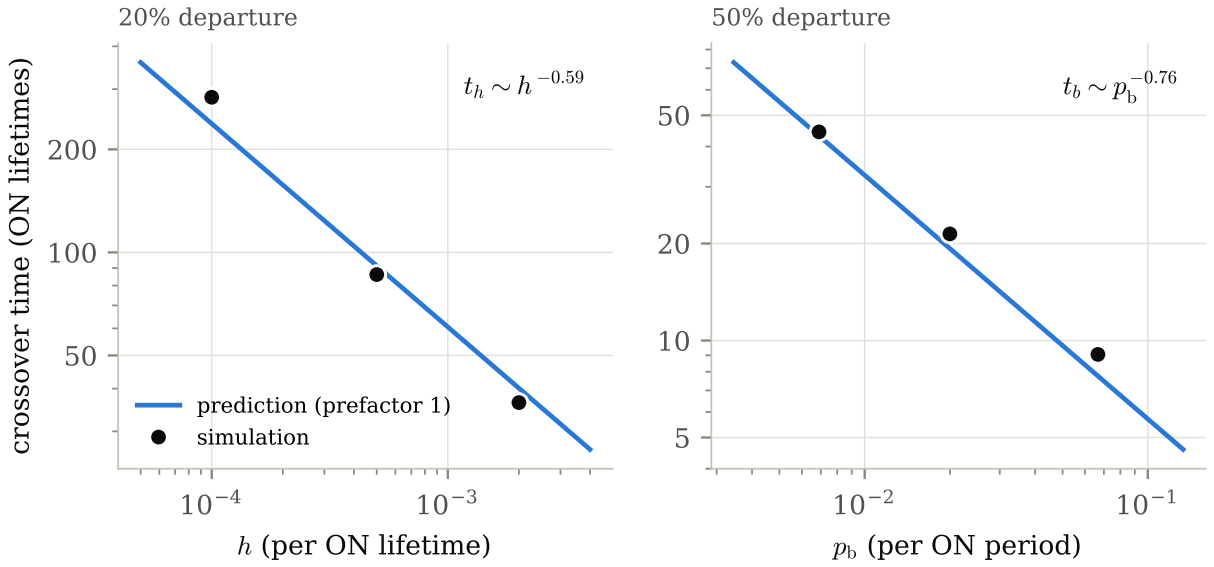


Figure 4: Crossover times against perturbation strength, log–log. Lines: the scaling predictions with unit prefactor; points: simulation.

5 Consequences for an experiment

1. **Scaling of the crossover, not of the decay.** The critical decay itself is visible over about one decade, too little to extract δ. The crossover times scale as $h^{-0.59}$ and $p_b^{-0.76}$, so varying the bleaching budget (oxygen scavenger, thiol) or the recovery rate and measuring the duration of the flash tests DP exponents without two decades of power law.

2. **An independent field.** Weak 405 nm illumination reactivates Cy5, tuning h without changing r. pH changes r through p_{off}; the two knobs together scan the (r, h) plane.

3. **What the pair experiment measures.** By Eq. (4), a pair measurement of $B(d)$ determines η/p_{off}, the single photophysical ratio that sets every coupling in the lattice.

4. **Testing κ^2.** Comparing pairs with rigid and flexible linkers, or measuring dye anisotropy on origami, decides whether orientation disorder must be included.

Caveats

The weak-transfer limit is used in Eq. (4); at short range ($d \lesssim 5$ nm) the full E_{ij} must be kept. The bleaching argument treats dilution as a uniform shift of the control parameter and ignores the spatial correlation of bleached sites, which follow past activity. Dyes are treated as two-state; triplet, *cis* and radical intermediates are folded into the rates.

References

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